# Week 27: Stable pairings

Adult guide built from the problem files · Unpiloted. The table kit is physically unrehearsed.

## Problems and routes

| Problem | K–1 | 2–3 | 4–5 | route |
|---|---|---|---|---|
| Which pairing can stay? | P1 | | | first |
| Draw every pairing that can stay | P2 | | | first |
| Swap one strip | P3 | | | return |
| Fill the empty strip | P4 | | | return |
| Only the left pairing | P5 | | | return |
| Both pairings stay | P6 | | | return |
| Circle the pairings that stay, join a blocker | | P1 | | first |
| Every pairing that can stay: one | | P2 | | first |
| Every pairing that can stay: three | | P3 | | first |
| Four pairs in two blocks | | P4 | | return |
| Exactly two that stay | | P5 | | return |
| All stable pairings | | | P1 | first |
| The asking rules, from both sides | | | P2 | first |
| A first-come rule | | | P3 | return |
| The rules always finish | | | P4 | return |
| The finish is stable | | | P5 | return |
| As many last choices as possible | | | P6 | return |

**First route.** K–1 P1–P2; Grades 2–3 P1–P3; Grades 4–5 P1–P2. **Return visits.** K–1 P3–P6; Grades 2–3 P4–P5; Grades 4–5 P3–P6.

## Key, one entry per problem

### Which pairing can stay?

K–1 P1 · first route

*Idea.* With two pairs, the pairing that can stay is the one with no blocker: in both cases only AX / BY stays, once because each letter has its first choice and once because A and X rank each other first.

### Draw every pairing that can stay

K–1 P2 · first route

*Idea.* A profile can have two stable pairings (top: AX / BY and AY / BX) or one (bottom: only AY / BX).

### Swap one strip

K–1 P3 · return route

*Idea.* One blocker can be removed by reversing either of its two strips (top: A's or Y's); when two blockers share no letter, no single strip repairs the pairing (bottom).

## Fill the empty strip

K–1 P4 · return route

*Idea.* Top: Y must rank A first for both pairings to stay. Bottom: impossible, because B and X block AX / BY whatever Y says.

## Only the left pairing

K–1 P5 · return route

*Idea.* Seven of the sixteen ways to fill four two-choice strips let only AX / BY stay; any two of them answer.

## Both pairings stay

K–1 P6 · return route

*Idea.* Exactly two of the sixteen ways let both pairings stay: the circles' first choices must be different squares and the squares' first choices the opposite circles.

## Circle the pairings that stay, join a blocker

Grades 2–3 P1 · first route

*Idea.* Top: only AX / BY stays, and A with X blocks AY / BX. Bottom: both pairings stay, although in each some letters wish for a change: a blocker needs a circle and a square who both want each other.

## Every pairing that can stay: one

Grades 2–3 P2 · first route

*Idea.* Of the six pairings of three circles and three squares, only AY / BZ / CX has no blocker.

## Every pairing that can stay: three

Grades 2–3 P3 · first route

*Idea.* Three of the six pairings stay (AY / BX / CZ, AZ / BX / CY, AZ / BY / CX); a complete answer names a blocker for each of the other three.

## Four pairs in two blocks

Grades 2–3 P4 · return route

*Idea.* A, B, W, X rank only each other ahead of the rest, and so do C, D, Y, Z; each block pairs stably in two ways, so exactly four of the 24 pairings stay, and none gives all eight letters their first choice.

## Exactly two that stay

Grades 2–3 P5 · return route

*Idea.* Lists can be designed so exactly two of the six pairings stay; explaining the rest means naming a blocker for each of the other four.

## All stable pairings

Grades 4–5 P1 · first route

*Idea.* The two-pair lists have two stable pairings, one giving the circles their first choices and one giving the squares theirs; the three-pair lists have exactly one, AY / BZ / CX.

## The asking rules, from both sides

Grades 4–5 P2 · first route

*Idea.* Deferred acceptance ends in a stable pairing, and the asking side can change it: circles asking give AW / BY / CZ / DX in 8 requests, squares asking give AZ / BY / CW / DX in 7.

## A first-come rule

Grades 4–5 P3 · return route

*Idea.* If receivers must keep their first asker, the run can end unstable: lists exist where a later asker and a receiver both prefer each other.

## The rules always finish

Grades 4–5 P4 · return route

*Idea.* No asker asks the same receiver twice, so four letters a side make at most 16 requests (ten make at most 100), and a receiver once asked stays held, so the run ends with everyone paired.

## The finish is stable

Grades 4–5 P5 · return route

*Idea.* If asker p prefers receiver r to its final partner, p asked r earlier and r released or rejected p for someone it prefers, and receivers only trade up, so r prefers its final partner: no blocking pair remains.

## As many last choices as possible

Grades 4–5 P6 · return route

*Idea.* In a stable pairing of three pairs at most three of the six letters get their last choice, and three is reached (checked over all 46,656 profiles).